

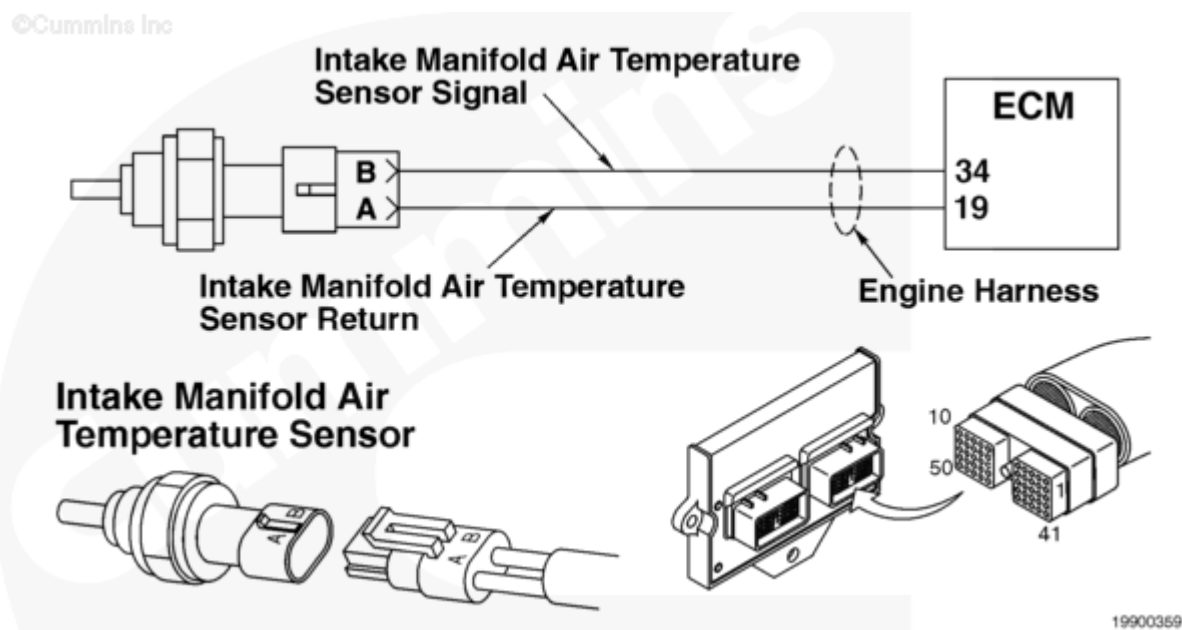
Fault Code: 154

Intake Manifold Temperature Sensor Circuit

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Overview

Codes	Reason	Effect
Fault Code: 154 PID(P): P105 SPN: FMI: 4 Lamp: Yellow SRT:	Less than 0.05 VDC detected at the intake manifold temperature sensor signal pin 34 of the engine harness electronic control module (ECM) connector.	Engine protection for the intake manifold air temperature is disabled.



Intake Manifold Temperature Sensor Circuit

Circuit Description

The intake manifold temperature sensor is used by the ECM to monitor the temperature of the air in the intake manifold downstream of the aftercooler element. The intake manifold temperature sensor is used by the ECM for the engine protection system and timing and fueling control.

The ECM monitors the voltage on pin 34. The ECM expects to see the voltage vary between 0.5 and 4.5 VDC. If the voltage is below 0.05 VDC, the ECM will log Fault Code 154.

Voltage below 0.05 VDC on pin 34 can be caused by short circuits to ground on the supply or return wires, or an internally grounded, failed sensor.

Component Location

Two intake manifold temperature sensors are used on the QST30 industrial engine, one on each side. The sensors are located in the intake manifold at the rear of the engine.

Shoptalk

The resistance of the sensor varies with the temperature. The reading you observe should compare to the following table if the sensor is functioning properly.

Temperature (°C)	Temperature [°F]	Resistance (ohms)
0	32	30k to 36k
25	77	9k to 11k
50	122	3k to 4k
75	167	1350 to 1500
100	212	600 to 675

Refer to Troubleshooting Fault Code t05-154 (/qs3/pubsys2/xml/en/procedures/87/87-t05-154.html)